

Arrays & Pointers

Characters & strings

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What we've learnt and today's plan

- Bitwise operations
- Pointers: they are addresses
 - With pointers arguments, a callee can modify local variables in the caller.
- Arrays: contiguous elements of the same type
 - Using pointers (w/ pointer arithmetic) to access array elements

Pass array to function via pointer

```
// multiply every array element by 2
void multiply2(int *a) {
    for (int i = 0; i < ???; i++) {
        a[i] *= 2;
    }
}
```

```
int main() {
    int a[2] = {1, 2};
    multiply2(a);
    for (int i = 0; i < 2; i++) {
        printf("a[%d]=%d", i, a[i]);
    }
}
```

Pass array to function via pointer

```
// multiply every array element by 2
void multiply2(int *a, int n) {
    for (int i = 0; i < n; i++) {
        a[i] *= 2; // (*(a+i)) *= 2;
    }
}
```

```
int main() {
    int a[2] = {1, 2};
    multiply2(a, 2);
    for (int i = 0; i < 2; i++) {
        printf("a[%d]=%d", i, a[i]);
    }
}
```

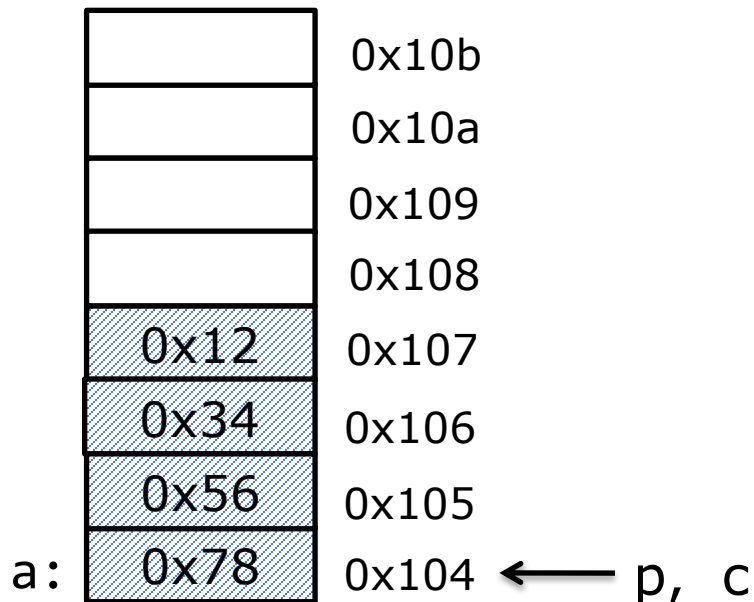
Pointer casting

```
int a = 0x12345678;  
int *p = &a;  
char *c = (char *)p;  
printf("%x\n", *c);
```

Output? (when running on Intel laptop)

Pointer casting

```
int a = 0x12345678;  
int *p = &a;  
char *c = (char *)p;
```

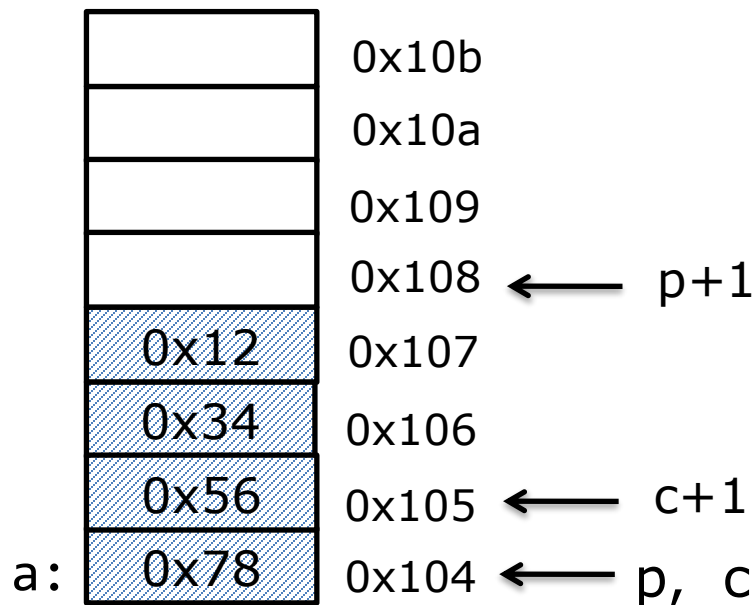


Intel laptop is small endian
*c is 0x78

What is `c+1`? `p+1`?

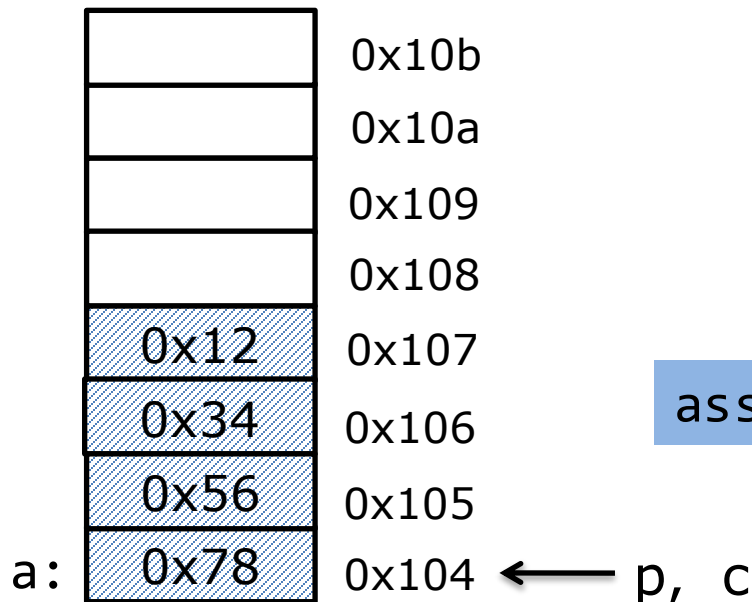
Pointer casting

```
int a = 0x12345678;  
int *p = &a;  
char *c = (char *)p;
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Pointer casting

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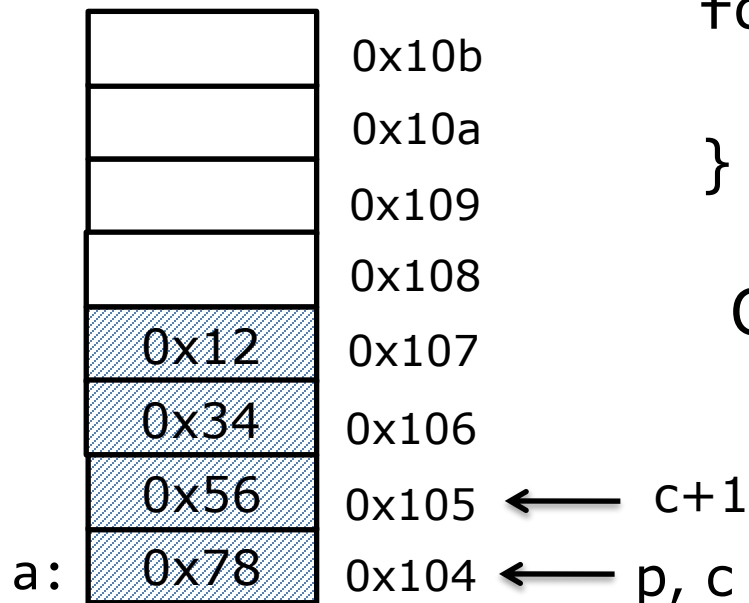


```
assert(p+i == (char *)p + i*sizeof(*p))
```

`sizeof(*p)`, or `sizeof(int)` is a C built-in that returns size of object/expression

Pointer casting

```
int a = 0x12345678;  
int *p = &a;  
char *c = (char *)p;
```



```
for (int i = 0; i < 4; i++) {  
    print("%x ", c[i]);  
}
```

Output: 0x78 0x56 0x34 0x12

What about big endian?

Characters

How to represent text characters?

- How to associate bit patterns to integers?
 - base 2
 - 2's complement
- How to associate bit patterns to floats?
 - IEEE floating point representation (based on normalized scientific notation)
- How to associate bit patterns to characters?
 - by convention
 - ASCII, UTF

ASCII: American Standard Code for Information Exchange

- Developed in 60s, based on the English alphabet
- use one byte (**with MSB=0**) to represent each character
- How many unique characters can be represented?

128

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

char

```
char c = 'A';
```

```
char c = 65;
```

```
char c = '\x41';
```

```
printf("c=%c\n", c);
```

```
printf("c=%d\n", c);
```

char

```
char c = 'A';  
c++;  
printf("c=%c, c=%d\n", c, c);
```



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C exercise 1: tolower

```
// tolower returns the corresponding
// lowercase character for c if c is an
// uppercase letter. Otherwise, it returns c.
char tolower(char c) {

    // test if c is an uppercase letter
    if (c < 'A' || c > 'Z') {
        return c;
    }

}
```

C exercise 1: tolower

```
// tolower returns the corresponding
// lowercase character for c if c is an
// uppercase letter. Otherwise, it returns c.
char tolower(char c) {

    // test if c is an uppercase letter
    if (c < 'A' || c > 'Z') {
        return c;
    }

    return c + ('a' - 'A');
}
```

C's standard library includes
tolower, toupper

C exercise 2: toDigit

```
// toDigit returns the corresponding integer for c
// assuming c is a valid digit character, e.g '1', '2',
int toDigit(char c) {
    // crash if c is not a valid character
    assert (c >= '0' && c <= '9');

    return c - '0';
}

int main() {
    int d = toDigit('8');
    printf("int is %d, multiply-by-2 %d\n", d, 2*d);
}
```

The Modern Standard: UniCode

- ASCII can only represent 128 characters
 - How about Chinese, Korean, all of the worlds languages? Symbols? Emojis?
- Unicode standard represents >135,000 characters

<u>U+1F600</u>		grinning face
<u>U+1F601</u>		beaming face with smiling eyes
<u>U+1F602</u>		face with tears of joy
<u>U+1F923</u>		rolling on the floor laughing
<u>U+1F603</u>		grinning face with big eyes

UTF-8

- UTF-8 is one encoding form for Unicode
 - use 1, 2, or 4 byte to represent a character
 - Unicode for ASCII characters have the same ASCII value → UTF-8 one byte code is the same as ASCII
- C has no primitive support for Unicode

C Strings

Strings

- String is represented as an array of chars.
 - Array has no space to encode its length.
- How to determine string length?
 - explicitly pass around an integer representing length

```
// tolower_string turns every character in character array s  
// into lower case
```

```
void tolower_string(char *s, int len) {  
    for (int i = 0; i < len; i++) {  
        s[i] = tolower(s[i]);  
    }  
}
```

Strings

- String is represented as an array of chars.
 - Array has no space to encode its length.
- How to determine string length?
 - explicitly pass around an integer representing length
 - C string stores a NULL character to mark the end (by convention)

```
void tolower_string(char *s) {  
  
}
```

Strings

- String is represented as an array of chars.
 - Array has no space to encode its length.
- How to determine string length?
 - explicitly pass around an integer representing length
 - C string stores a NULL character to mark the end (by convention)

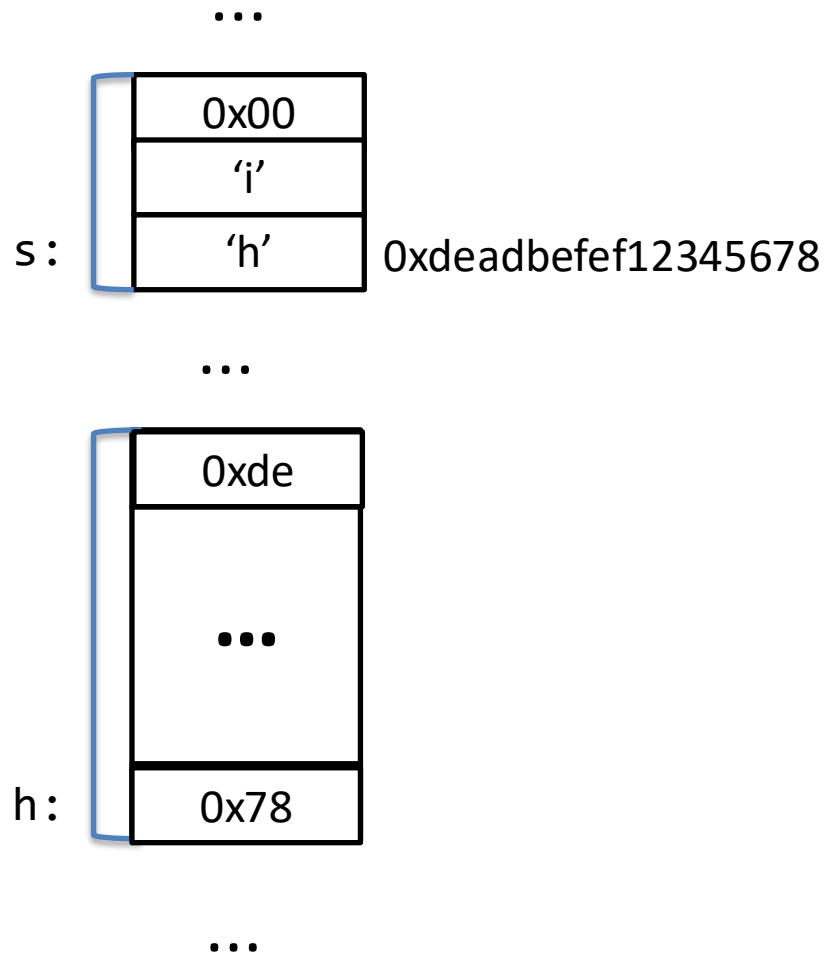
```
void tolower_string(char *s) {  
    int i = 0;  
    while (s[i] != '\0') {  
        s[i] = tolower(s[i]);  
        i++;  
    }  
}
```

Copying string?

does this make a copy of "hi"?

```
char s[3] = {'h', 'i', '\0'};  
char *h;  
h = s;  
h[0] = 'H';
```

```
printf("s=%s h=%s\n", s, h);
```



Copying string?

does this make a copy of “hi”?

```
char s[3] = {'h', 'i', '\0'};
```

```
char h[3];
```

```
h = s;
```

```
h[0] = 'H';
```

```
printf("s=%s h=%s\n", s, h);
```

Copying string

```
void strcpy(char *dst, char *src)
{

}
```

```
int main()
{
    char s[3] = {'h', 'i', '\0'};
    char h[3];
    strcpy(h, s);
    h[0] = 'H';

    printf("s=%s h=%s\n", s, h);
}
```

Copying string

```
void strcpy(char *dst, char *src) {  
    int i = 0;  
    while (src[i] != '\0') {  
        dst[i] = src[i];  
        i++;  
    }  
}
```

strcpy is included in C std library.

```
int main() {  
    char s[3] = {'h', 'i', '\0'};  
    char h[3];  
    strcpy(h, s);  
    h[0] = 'H';  
  
    printf("s=%s h=%s\n", s, h);  
}
```

Copying string

```
void strcpy(char *dst, char *src) {  
    int i = 0;  
    while (src[i] != '\0') {  
        dst[i] = src[i];  
        i++;  
    }  
}
```

```
int main() {  
    char s[3] = {'h', 'i', '\0'};  
    char h[2];  
    strcpy(h, s);  
    h[0] = 'H';
```

Results in out-of-bound write!
Buffer overflow!

```
    printf("s=%s h=%s\n", s, h);  
}
```

Copying string

```
void strncpy(char *dst, char *src, int n) {  
    int i = 0;  
    while (src[i] != '\0' && i < n) {  
        dst[i] = src[i];  
        i++;  
    }  
}
```

strncpy is included in C std library.

```
int main() {  
    char s[3] = {'h', 'i', '\0'};  
    char h[2];  
    strncpy(h, s, 2);  
    h[0] = 'H';  
  
    printf("s=%s h=%s\n", s, h);  
}
```

A different way of initializing string

...

```
char s1[3] = {'h', 'i', '\0'};
```

```
//equivalent to
```

```
//char s1[3] = "hi";
```

```
char *s2 = "bye";
```

```
s1[0] = 'H';
```

← OK

```
s2[0] = 'B';
```

← Segmentation fault (bus error)

```
printf("s1=%s s2=%s\n", s1, s2);
```

A different way of initializing string

```
char s1[3] = {'h', 'i', '\0'};
```

```
//equivalent to
```

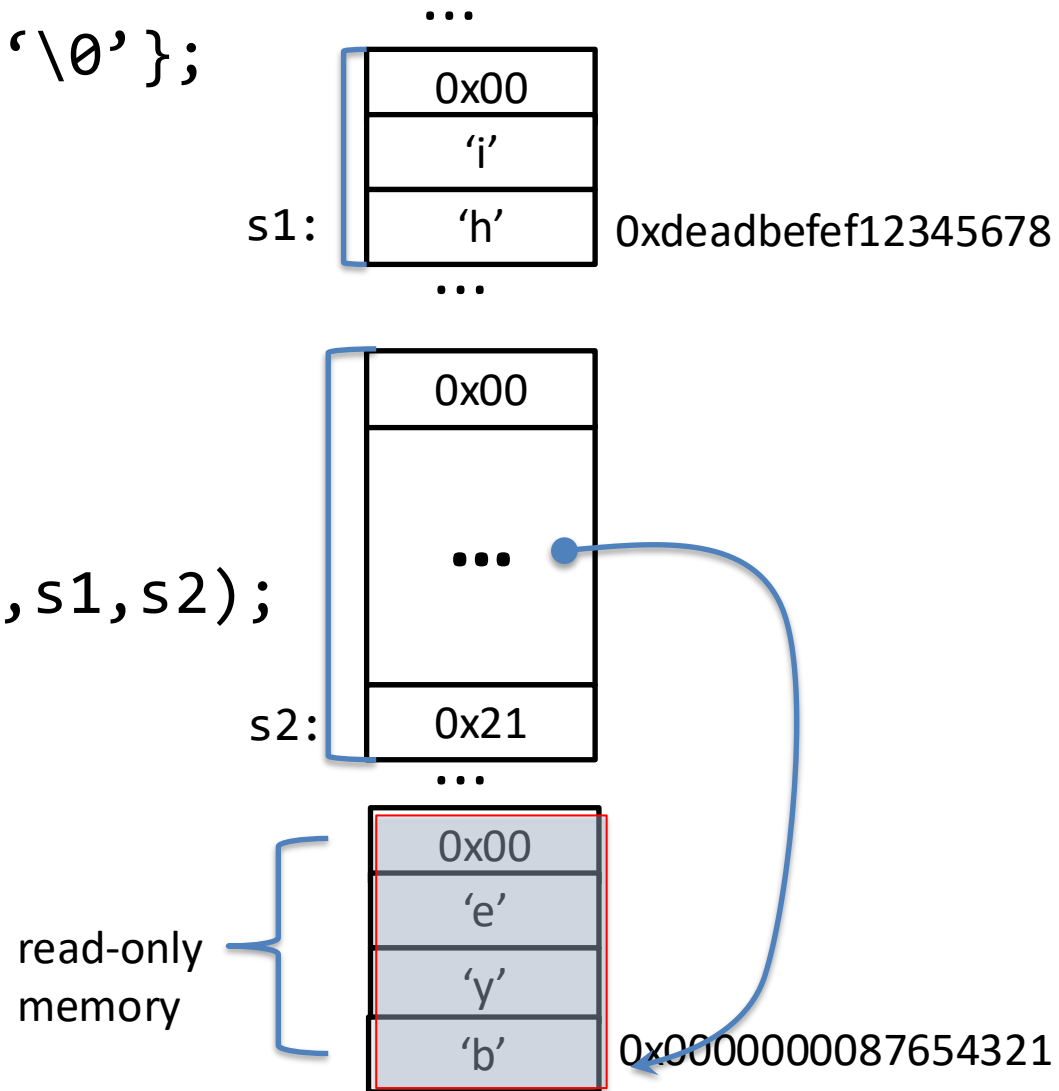
```
//char s1[3] = "hi";
```

```
char *s2 = "bye";
```

```
s1[0] = 'H';
```

```
s2[0] = 'B';
```

```
printf("s1=%s s2=%s\n", s1, s2);
```



The Atoi function

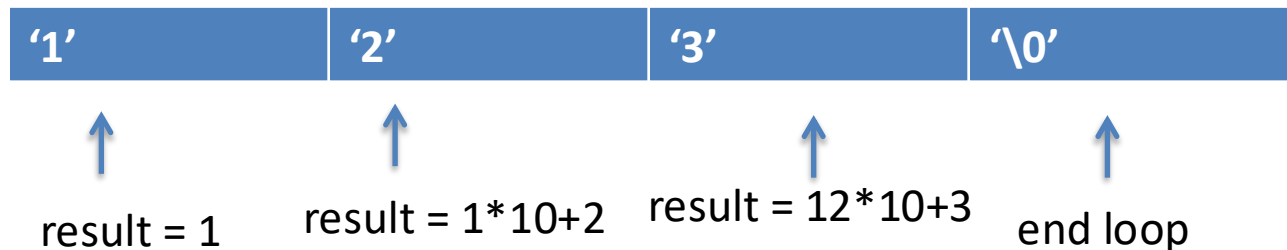
```
// atoi returns the integer
// corresponding to the string of digits
int atoi(char *s)
{

}

int main()
{
    char *s= "123";
    printf("integer is %d\n", atoi(s));
}
```

The Atoi function

```
// atoi returns the integer
// corresponding to the string of digits
int atoi(char *s) {
    int result = 0;
    int i = 0;
    while (s[i] >= '0' && s[i] <= '9') {
        result = result * 10 + (s[i] - '0');
        i++;
    }
    return result;
}
```

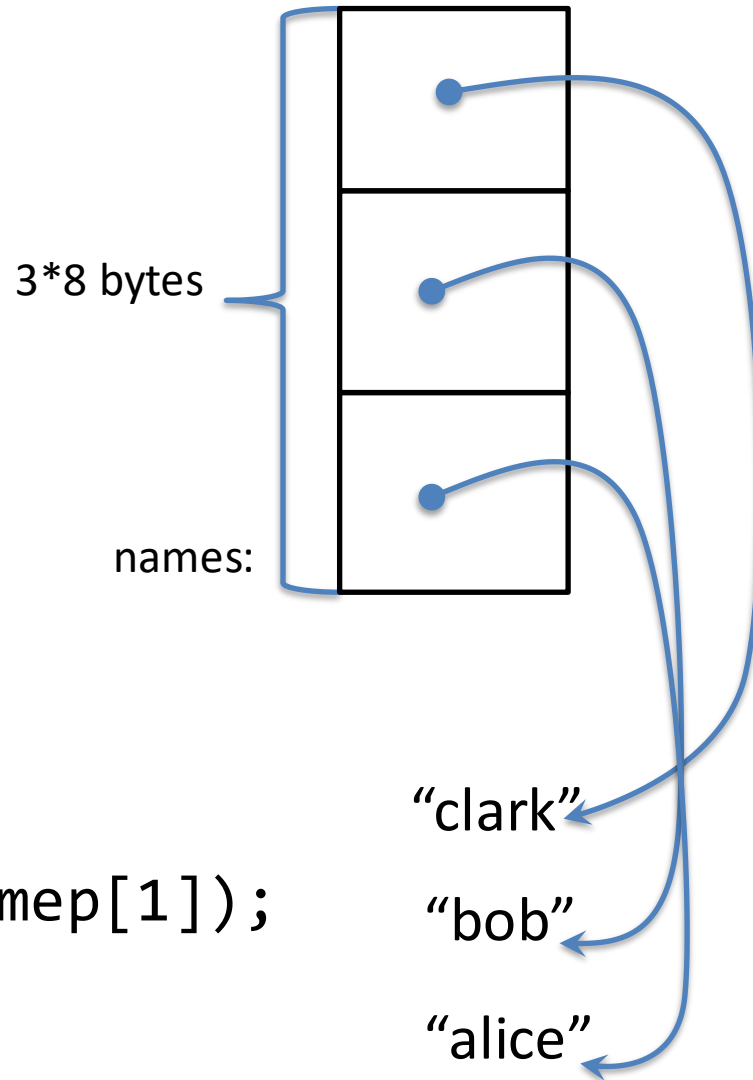


Array of pointers

```
char* names[3] = {  
    "alice",  
    "bob",  
    "clark"  
};
```

```
char **namep;  
namep = names;
```

```
printf("name is %s", namep[1]);
```



The most commonly used array of pointers: argv

```
int main(int argc, char **argv)
{
    for (int i = 0; i < argc; i++) {
        printf("%s\n", argv[i]);
    }
}
```

```
$ ./a.out 1 2 3
./a.out 1 2 3
```



argv[0] is the name of the executable